

## Arduino UNO + Measuring Ux, Rx and Cx

My last idea was to create measuring system with Arduino Uno due its inputs A0 to A7 or A0 to A3 or for A0 only.

So I've programed a sketch for Arduino Uno or Nano , the last version is now `Arduino_mesurer_URC_en_3.ino`

I've used in it my command system for serial communication via com ports from PC similar of my previous ones with command letters A to ... V at last (so X,Y and Z are still free ... )

For example A command call the measuring of U on input A0 and return the value as integer number for this example max value 1023 in form \*A,1023,#,

B command call measuring from inputs A0 to A3 ,

C command call measuring from inputs A0 to A7

I've programmed two programs with my Delphi XE Starter, at first it was program SerialReader last version 15 suitable for reading data from my different systems with Arduinos and ESP microcomputers and next the program SerialMeasurer last version 11 specialised for a measuring of Ux in ranges 0 to 5 or 10 Volts or  $\pm 3V$ , Rx and Cx .

I've created at first specialised module for measuring of U in Volts. There is my simpler schema for this type for my purpose :

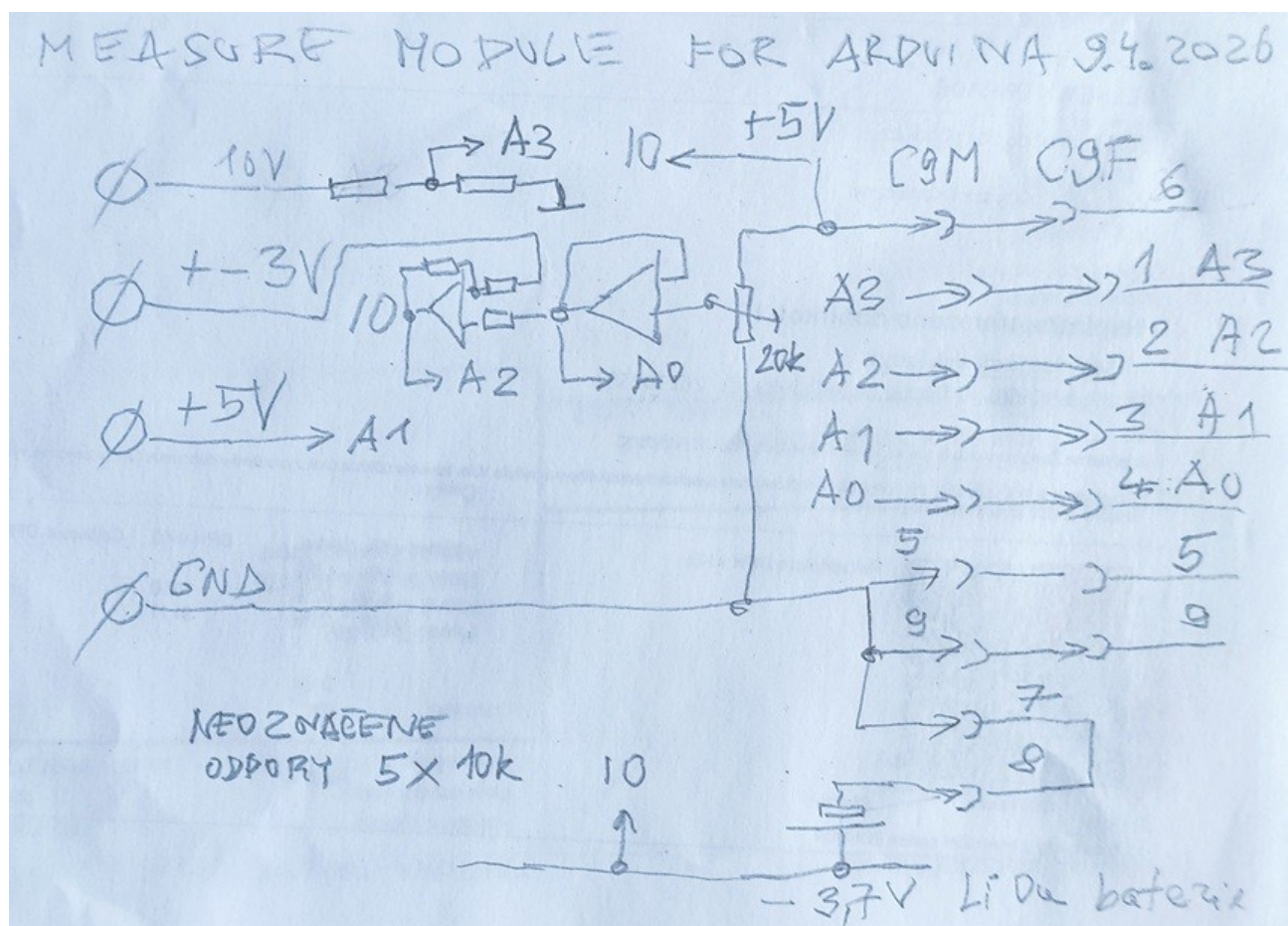


Fig. 1 - the basic principal schema of measuting U in Volts with Arduino Uno

and there is photo of this U specialised module on the next page :

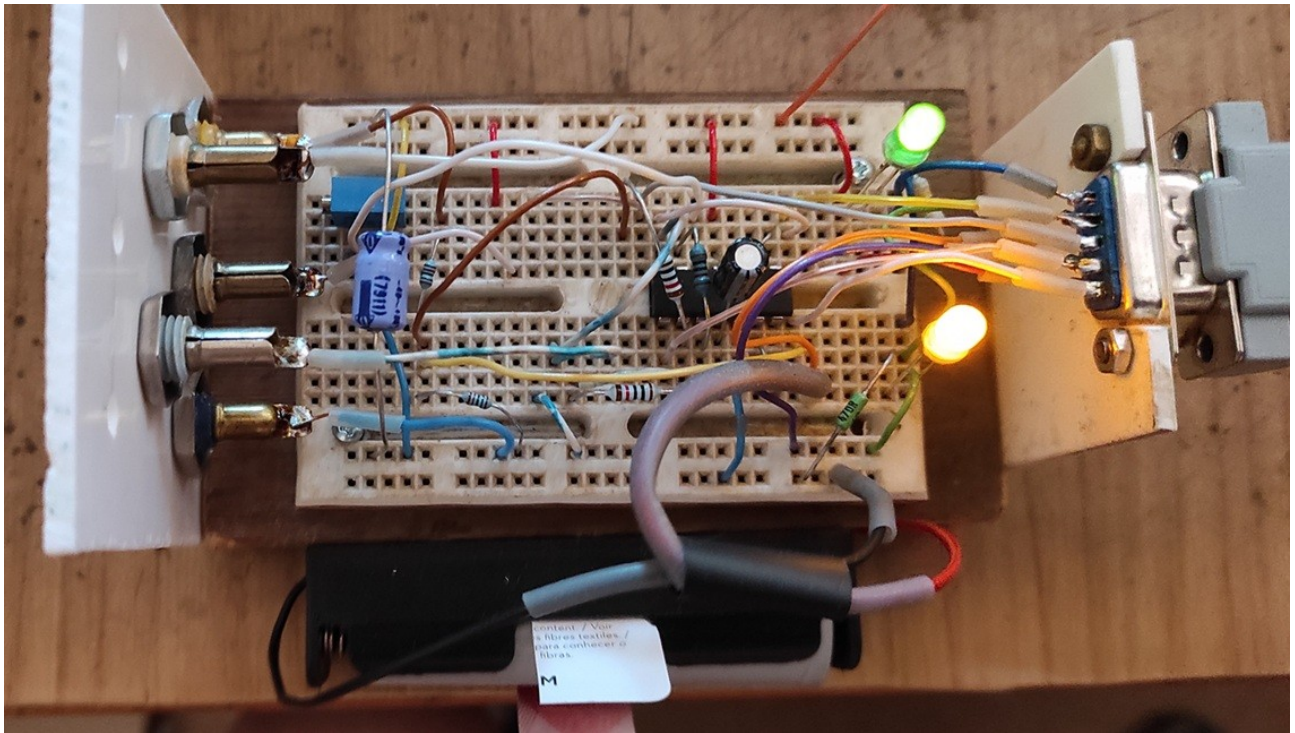


Photo 1 - U module for measuring Volts + 5 V, + 10 V and for measuring Volts +- 3 V cca

So I've created the next similar simpler module for measurig U , too and Rx resistor or Cx capacitors. There is my basic principal schema for this URC module :

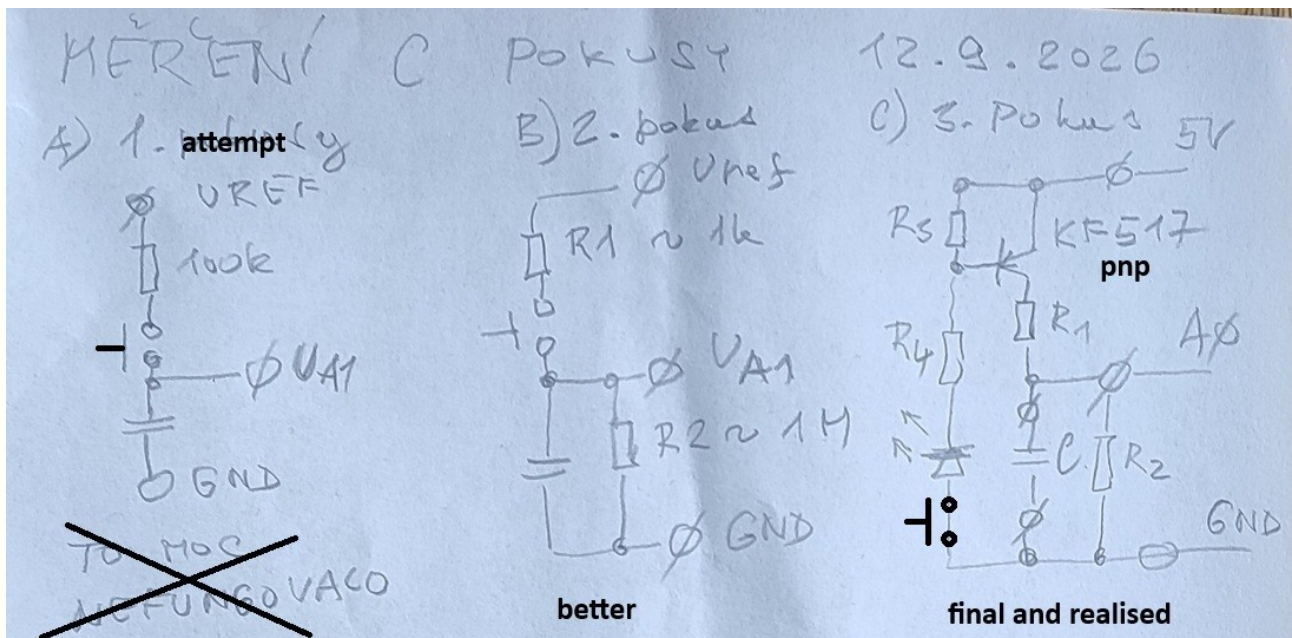


Fig. 2 - the basic principal schema of module for measuring U integer nr., Rx in kilo Ohm and Cx in nano till micro, maybe mili Farads

There is photo of my URC module for Arduino Uno (what gives better +5 V from USB coupling on PC as Nano, where is cca 4.8 V only) with my description on the next page :

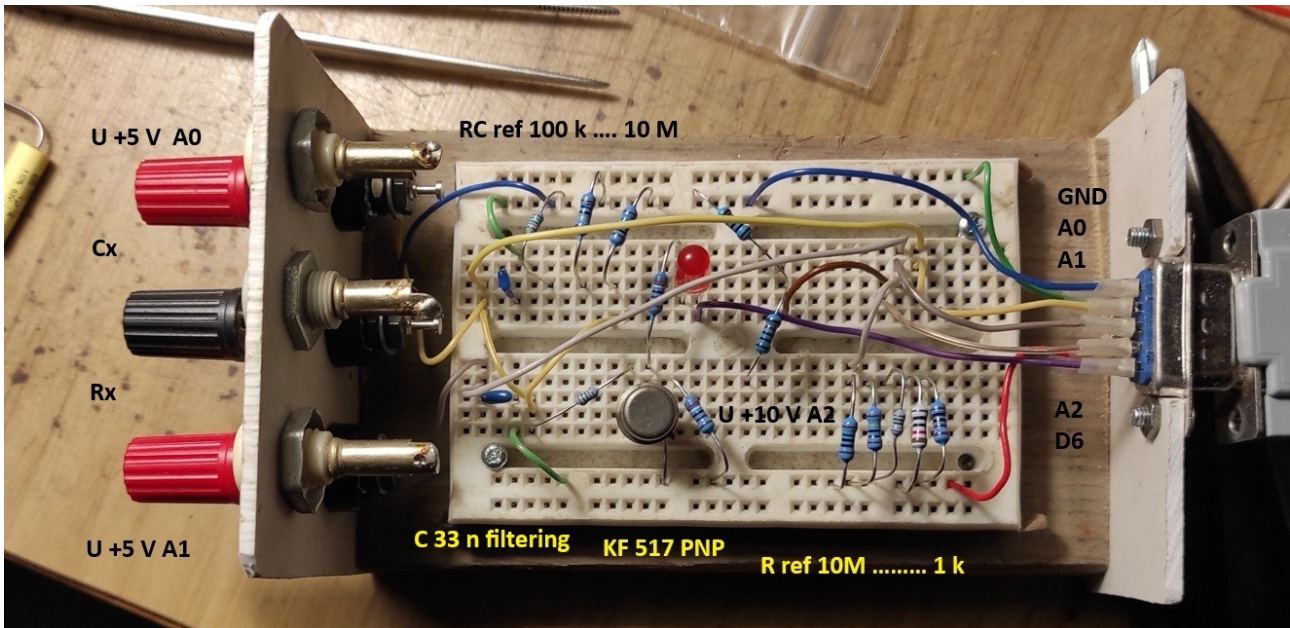


Photo 2 - URC module for measuring U, Rx and Cx with Arduino Uno

There is possible with this URC module to meas Rx from kilo Ohms to about 10 M Ohms and Cx capacitance from nano Farads to mili Farads, too.

I know there is another possibility how to meas capacitance smaller as nano Farads with Arduino, too. There is on the next figure of the basic principal schema of CxNf module:

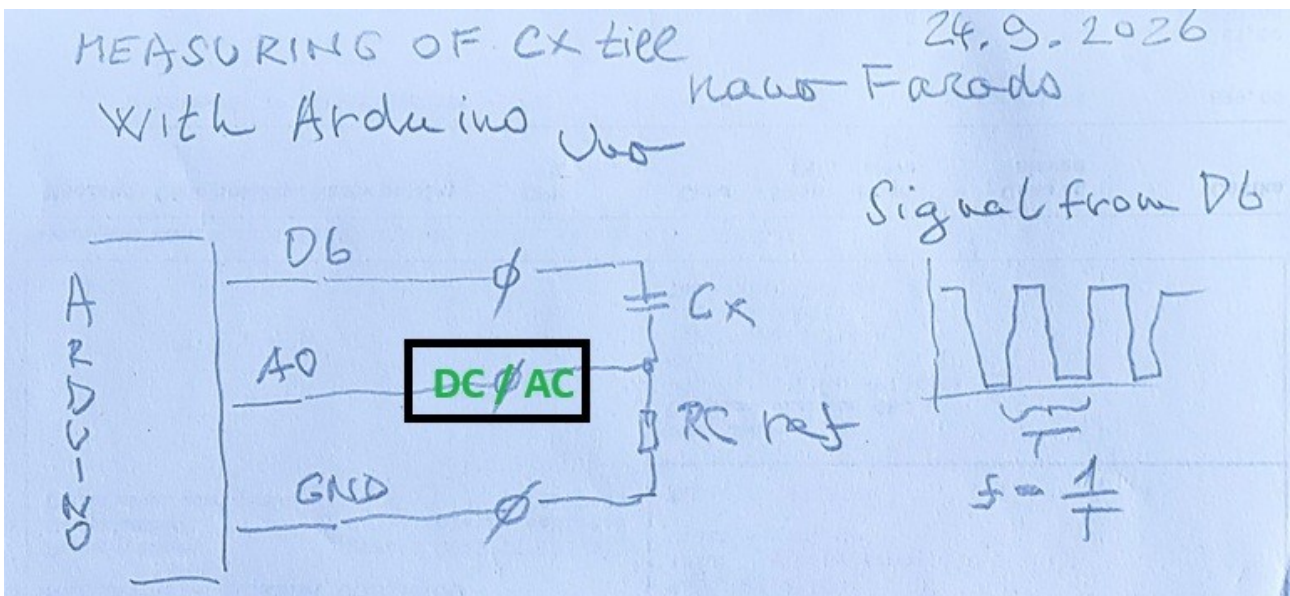


Fig. 3 - the basic principal schema of CxNf module for the next attempts and tests

So I am interested to re-arrange my URC module or to create new one for this type of measuring Cx with smaller capacitance as 10 nf, what is with with my URC module on the end of its possibilities.

Instead of it I must re-arrange my program for URC measuring for this Nf system. There is printscreen of window from SerialMeasurer11 specialised for URC measuring :

Form Modbus Display RC

File:  Task

Device:  Descript:

Task file:

Task Actual:  Inputs all:  Outputs:

Inputs selected:

Calibrated output:  ☐ Calibration

Calibration parameters:  Type:  Range:  Zero:  Kal value top:  bottom:

Time abs:  rel:  nr:  lcs:  Interval meas.:  Com:

8 [V] off off off off

U2.5Vref(A0) off off off

399 399

Meas RC ☐ Data Record ☐ R x ☒ C x  Calib Tau:

Set Uref Set U Rx R ref:  C in:  Limit[sec]:  Count R in C

U ref: Count Rx ☐ Test R ref Count Cx Tau [%]:  R (in C)

930 400 22.943

Progres:   (100) Status:  

Fig. 4 - Window from SerialMeasurer11 specialised for URC measuring with example of measuring Cx 22 micro Farads.

I think and see there is no space for the next Nf measuring , so it is task for my next project.

#### Notes:

First - the accuracy of measuring in my realised Cx system is coupled on the set of expected time of  $\tau = R_{ref} \times C_x$  and in the theory it is the counted constant 36.8 % cca, what is possible to correct by setting of this constant in percent and in this example I correct this one on 43% for the calibration of my result with my non calibrated Cx 22 M capacitor.

Second - In my program I've counted Arduino inputs from A1 to A7 as 1 to 7, so the input A0 is in my notation marked with number 8 ... I did it because in my previous programs were my inputs counted from 1 to 8 always.

Third - my project to meas the internal Rx in Cx capacitor was my next attempt what is not ended and I thing it is complicated idea. But it can be solved when you will have long time for measure of how long will be your capacitor re-charged or with permanent coupling on A0 input of Arduino or if you will meas only samples of Volts after some elapsed time.

Fourth - Form Modbus Display RC in the tittle row of window there is Modbus only rest from my previous programs for another purpose with Modbus serial communication.

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